

Should the Digital Euro Bear Interest?

Programmable holding costs, the lower bound,
and why the only usable form is demurrage

[Presenter name]

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Two valves on the digital euro

Quantity valve: holding limit

- Council and Parliament both want a cap
- Range under discussion:
€1,500–3,000
- Trilogue ongoing (3rd round 30 Sep 2026)
- → companion talk

Price valve: remuneration

- Commission proposal, Art. 16(8):
“The digital euro shall not bear interest.”
- The price valve is **welded shut at zero**

**What does closing the price valve cost,
and what would opening it require?**

Three claims

1. **An unremunerated digital euro hardens the lower bound.**

It is cash without storage costs: on capped balances the deposit-rate floor moves from $-c_{\text{cash}}$ back to 0. (Known in policy work; I give a one-line formalisation.)

2. **If a holding cost is programmed, it must be demurrage, not expiry.**

Expiry is a fiscal-programme tool; only continuous decay is an interest rate. (Model result.)

3. **Remuneration flexibility protects policy space; it does not create it.**

While cash exists, a negative digital-euro rate cannot go below the cash floor. Policy: keep zero as the default, legislate a contingent rate clause.

Why programmable money tempts policymakers

The macro problem

- Liquidity trap: $MV = PY$, but V falls when M rises
- Rates stuck at the effective lower bound (ECB DFR -0.50% , 2019–22)
- Desire to bypass banks and push spending directly

Expiry works, empirically

- China digital coupons: **¥3** extra spending per **¥1** of subsidy
Xing et al. (AEJ:Macro 2023); Ding et al. (2024): 3.1–3.3
- France RCT: one-month MPC **61%** with a 3-week expiry vs **23%** without
Boehm, Fize & Jaravel (2024)
- Japan 1999 coupons: weak effect
Hsieh, Shimizutani & Hori (JPubE 2010)

Question: can the digital euro itself carry this “use it or lose it” force?

What is already known

Strand	Key references	Status
Carry tax on money	Gesell (1916), Fisher (1933), Keynes (1936), Goodfriend (2000), Buiter & Panigirtzoglou (2003)	settled idea
Interest-bearing CBDC removes ZLB	Bordo & Levin (2017), Agarwal & Kimball (2015, 2019), Assenmacher & Krogstrup (2018)	settled idea
Tiered CBDC remuneration Zero-rate CBDC hardens ELB	Bindseil (2020), Bindseil & Panetta (2020) ECB Report on a digital euro (2020), Pirgmann & Wawrosz (2024)	ECB design input recognised
Expiring CBDC	Kahn, van Oordt & Zhu (JMCB 2026): expiry for <i>loss recovery</i>	different purpose
Expiring vouchers	Xing et al. (2023), Ding et al. (2024), Boehm et al. (2024)	empirical

Gap: no common framework comparing *expiry* and *demurrage* as two schedules of the same holding cost, and no mapping from voucher-strength stimulus to equivalent interest rates.

Central-bank money is the deposit's outside option

A depositor can always switch to a central-bank liability j with rate i_j and holding cost c_j . Banks cannot push the deposit rate below the best outside option:

$$r_D \geq \max_j \{ i_j - c_j \} \quad (\text{on balances that option can absorb})$$

Outside option	i_j	c_j	Floor on r_D
Cash	0	$c_{\text{cash}} > 0$ (storage, insurance)	$-c_{\text{cash}}$
Digital euro, Art. 16(8)	0	≈ 0	0, up to the limit $\bar{M}$
Digital euro at $-\lambda$	$-\lambda$	≈ 0	$\max\{-\lambda, -c_{\text{cash}}\}$

- The **ZLB** and **disintermediation** are the same mechanism seen from two sides
- The holding limit $\bar{M}$ decides *how much* money enjoys the zero floor (companion talk)

Claim 1: a zero-rate digital euro is frictionless cash

- 2019–22: the ECB went to -0.50% without a run into cash, because hoarding notes is costly
- A digital euro at 0 removes that cost on the first $\bar{M}$ per person
- Consequences when rates are negative:
 - deposits up to $\bar{M}$ flow out exactly when banks are weakest
 - pass-through of negative rates to retail balances is capped at 0
 - disintermediation pressure peaks in the **negative-rate** regime, not in normal times
- So zero remuneration and the holding limit are a **package**: the limit is what makes the zero floor affordable

Consistent with ECB (2020) and Pirgmann & Wawrosz (2024). New here: the floor formula and the link to $\bar{M}$.

Claim 2: a model of programmable holding costs

- Token value per unit: V . Useful purchases are worth 1; holders search with intensity κ at cost $\kappa^2/2a$
- A **junk outlet** (low-value goods, dumping on a secondary market) is always available at $\phi < 1$
- Discount rate r ; velocity is κ

Hard expiry at T ($\tau = T - t$)

$$\frac{dV}{d\tau} = -rV + \frac{a}{2}(1 - V)^2, \quad V(0) = \phi$$

Demurrage at rate λ

$$(r + \lambda)V = \frac{a}{2}(1 - V)^2$$

Optimal velocity in both cases: $\kappa^* = a(1 - V)$

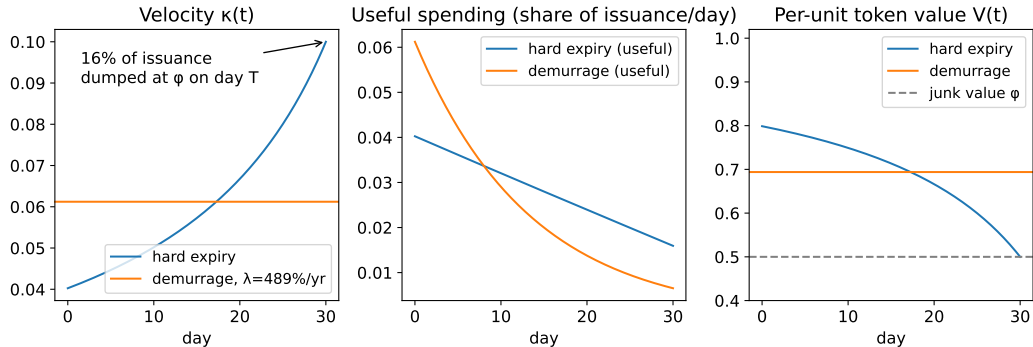
Result 1: velocity follows a square-root rule

$$\kappa^* \approx \sqrt{2a(r + \lambda)} \quad \Rightarrow \quad \frac{\partial \ln \kappa^*}{\partial \ln(r + \lambda)} = \frac{1}{2}$$

Demurrage λ (p.a.)	0.5%	1%	2%	3%	5%
Velocity multiplier ($r = 4\%$)	1.06	1.12	1.22	1.32	1.49

- Same elasticity as Baumol–Tobin money demand: the holding cost is just an interest rate
- Policy-sized rates move velocity **smoothly** by 10–30%: no deadline, no spike

Result 2: paths at equal average velocity ($T = 30$ days)



Expiry: velocity rises towards T , then **16%** of issuance is dumped at ϕ . Demurrage: constant velocity, constant value.

Result 3: expiry front-loads more, but wastes and fragments

T	Hard expiry				Demurrage (same avg. velocity)		
	useful	dumped	waste	value range	λ p.a.	useful	burned
7 d	45%	55%	27%	0.50–0.63	1172%	41%	15%
14 d	66%	35%	17%	0.50–0.71	842%	58%	17%
30 d	84%	16%	8%	0.50–0.80	489%	73%	16%
90 d	97%	3%	2%	0.50–0.91	161%	88%	10%

$a = 0.2$, $\phi = 0.5$, $r = 4\%$. “waste” = $(1 - \phi) \times \text{dumped}$ is a real resource loss; “burned” is a transfer to the issuer.

- **Correction to the naive story:** expiry delivers *more* useful spending within T
- Its costs lie elsewhere: real waste, and token value that depends on the vintage

Why programmed holding costs must take the form of demurrage

Property	Hard expiry	Demurrage
Real waste (junk spending at T)	2–27% of issuance	none if $V > \phi$
Value of one unit	depends on vintage	identical for all units
Fungibility / singleness of money	broken	preserved
Velocity path	rising, mass point at T	constant
Applicable to the whole money stock	no (needs vintages)	yes
Is it an interest rate?	no	yes: $i = -\lambda$

Expiry is a fiscal-programme tool. Demurrage is a monetary-policy instrument.

A central bank that programs a holding cost into money can only use the second.

Orders of magnitude: vouchers are not rates

- Matching voucher-strength velocity needs demurrage of **160%–1,200% per year**
- Policy-sized rates (-0.5% to -3%) raise velocity by **6–32%**
- Gap of **two to three orders of magnitude**
- Voucher effects also come from channels a rate does not have:
 - transfer of new money (income effect)
 - salience of a deadline (Boehm et al. 2024)
 - minimum-spend notches (Ding et al. 2024)

The digital euro cannot be a stimulus voucher. At most it can carry a policy rate.

Claim 3: can a negative digital-euro rate break the ELB?

Floor on deposit rates, with cash still in circulation:

$$\underbrace{\max\{0, -c_{\text{cash}}\} = 0}_{\text{Art. 16(8)}} \quad \text{vs} \quad \underbrace{\max\{-\lambda, -c_{\text{cash}}\} \geq -c_{\text{cash}}}_{\text{remunerated}}$$

- **Asymmetry:** zero remuneration *raises* the floor from $-c_{\text{cash}}$ to 0; negative remuneration *cannot* push it below $-c_{\text{cash}}$
- Going deeper needs cash reform: a crawling cash–reserve exchange rate (Agarwal & Kimball 2019)
- Other leaks: stablecoins, foreign currency, banks that do not pass rates on

Remuneration flexibility protects existing policy space; it does not create new space.

Policy proposal (for discussion)

Keep zero as the default, add a contingent rate clause

- Normal times: $i_{DE} = 0$, as in Art. 16(8) (trust, simplicity, no “shrinking money”)
- If the deposit facility rate is below zero, holdings above a tier threshold $M_1 < \bar{M}$ pay

$$i_{DE} = \max\{\text{DFR}, -\bar{c}\}, \quad \bar{c} \approx \text{cash holding cost}$$

- Implemented as **continuous demurrage**, never as expiry
- Tier 1 ($\leq M_1$) protects households and the “money” role
- Tier 2 removes the frictionless-cash arbitrage at the margin
- Rate bounded by $-\bar{c}$: no incentive to flee into banknotes

Takeaways

1. A zero-rate digital euro is **cash without storage costs**: it hardens the lower bound on the balances the limit allows
2. Programmed holding costs must be **demurrage**: expiry wastes resources, breaks fungibility and is not a rate
3. Vouchers and rates differ by **two to three orders of magnitude**: the digital euro cannot be a stimulus voucher
4. Remuneration flexibility **protects** policy space; only cash reform **creates** it

Companion talk: how large $\bar{M}$ should be. This talk: what price should sit under $\bar{M}$.

Backup

B1. Derivations

Holder's problem. Per unit of balance, with search cost $\kappa^2/2a$ and value V :

$$(r + \lambda)V = \max_{\kappa} \left\{ \kappa(1 - V) - \frac{\kappa^2}{2a} \right\} \Rightarrow \kappa^* = a(1 - V), \quad (r + \lambda)V = \frac{a}{2}(1 - V)^2.$$

Square-root rule. Let $x = 1 - V$ (small when $r + \lambda \ll a$):

$$(r + \lambda)(1 - x) = \frac{a}{2}x^2 \Rightarrow x \approx \sqrt{2(r + \lambda)/a}, \text{ so } \kappa^* = ax \approx \sqrt{2a(r + \lambda)}.$$

Expiry. With remaining time τ , the HJB becomes the ODE $dV/d\tau = -rV + \frac{a}{2}(1 - V)^2$, $V(0) = \phi$. V rises above ϕ immediately if $\frac{a}{2}(1 - \phi)^2 > r\phi$, so dumping happens only at T . Velocity $\kappa(\tau) = a(1 - V(\tau)) \uparrow a(1 - \phi)$ as $\tau \rightarrow 0$: **finite**, but the remaining balance is a mass point at T .

Secondary-market discount. An expiring unit trades at $V(\tau)$, so the discount $1 - V(\tau)$ rises from about $1 - V(T)$ to $1 - \phi$; units of different vintages trade at different prices.

Calibration. λ is chosen so that $\kappa_{\text{demurrage}}$ equals the time-average of $\kappa_{\text{expiry}}(t)$ over $[0, T]$.

B2. Sensitivity ($T = 30$ days)

ϕ	a	useful	dumped	waste	value range	λ p.a.	useful	burned
0.3	0.2	90%	10%	7%	0.30–0.77	833%	73%	22%
0.5	0.2	84%	16%	8%	0.50–0.80	489%	73%	16%
0.7	0.2	72%	28%	8%	0.70–0.84	211%	68%	9%
0.5	0.1	67%	33%	16%	0.50–0.71	403%	59%	17%
0.5	0.4	94%	6%	3%	0.50–0.87	506%	83%	12%

Robust across parameters: expiry front-loads more, always wastes resources and always creates vintage-dependent prices; the matching demurrage rate is always in the hundreds of percent per year.

Code: `model/expiry_vs_demurrage.py`

B3. Limitations

- Partial equilibrium: no prices, no general-equilibrium feedback from extra spending
- Search technology is stylised; a and ϕ are not estimated
- The “cliff after T ” is not modelled; evidence is mixed: Xing et al. (2023) find purchases are moved forward, Ding et al. (2024) find no short-run intertemporal substitution
- Behavioural channels (salience, notches) are outside the model and favour vouchers
- c_{cash} is not estimated; the ELB literature gives a range rather than a number

B4. References

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